STA 2002, Summer 2024
Probability and Statistics II

Contingency Tables

Motivation: Test for Homogeneity

- Two independent experiments, each experiment has *k* mutually exclusive and exhaustive outcomes, $A_1, A_2, \ldots, A_k$

	A_1	A_2	A_3	$\cdots$	A_k
First experiment	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$\cdots$	$p_{1,k}$
Second experiment	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$\cdots$	$p_{2,k}$

Two multinomial distributions

$$p_{1,1} + p_{1,2} + \cdots p_{1,k} = 1$$

$$p_{2,1} + p_{2,2} + \cdots p_{2,k} = 1$$

- Test homogeneity (the corresponding probabilities associated with the two independent experiments are equal)

$$H_0: \ p_{1,1} = p_{2,1}, \ p_{1,2} = p_{2,2}, \ p_{1,3} = p_{2,3}, \ldots, \ p_{1,k} = p_{2,k}$$

Reading: Chapter 9.2

Recap: Chi-square test (one-sample)

Data

$$Y_1, \ldots, Y_n \sim \text{Multinomial}(n, p_1, \ldots, p_k)$$

Null Hypothesis

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \ldots, k,$$

(Chi-square) Test Statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(y_i - np_{i0})^2}{np_{i0}}$$

Distribution under H_0

$$\chi^2 \sim \chi^2(k-1)$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha(k-1)$$

at significance level α

p-value $P(\chi^2_\alpha(k-1) > \chi^2)$

Chi-square test for Homogeneity

	A_1	A_2	A_3	$\cdots$	A_k
First experiment (repeat n_1 times)	Observed: $Y_{1,1}$ Expected: $n_1 p_{1,1}$	Observed: $Y_{1,2}$ Expected: $n_1 p_{1,2}$	Observed: $Y_{1,3}$ Expected: $n_1 p_{1,3}$	$\cdots$	Observed: $Y_{1,k}$ Expected: $n_1 p_{1,k}$
Second experiment (repeat n_2 times)	Observed: $Y_{2,1}$ Expected: $n_2 p_{2,1}$	Observed: $Y_{2,2}$ Expected: $n_2 p_{2,2}$	Observed: $Y_{2,3}$ Expected: $n_2 p_{2,3}$	$\cdots$	Observed: $Y_{2,k}$ Expected: $n_2 p_{2,k}$

- From the chi-square test, we know

$$\sum_{j=1}^{k} \frac{\left(y_{1,j} - n_1 p_{1,j}\right)^2}{n_1 p_{1,j}} \sim \chi^2(\boldsymbol{k-1}) \qquad \sum_{j=1}^{k} \frac{\left(y_{2,j} - n_2 p_{2,j}\right)^2}{n_2 p_{2,j}} \sim \chi^2(\boldsymbol{k-1})$$

$$\Rightarrow \quad \sum_{i=1}^{2} \sum_{j=1}^{k} \frac{\left(y_{i,j} - n_i p_{i,j}\right)^2}{n_i p_{i,j}} \sim \chi^2\left(\boldsymbol{2(k-1)}\right)$$

Chi-square test for Homogeneity

- However, if $p_{i,j}, i = 1,2, j = 1, \dots, k$ are **unknown**, under H_0, we estimate $p_1, p_2, \dots, p_k$

$$H_0: \ p_{1,1} = p_{2,1} = \boldsymbol{p_1}, \ p_{1,2} = p_{2,2} = \boldsymbol{p_2}, \dots, \ p_{1,k} = p_{2,k} = \boldsymbol{p_k}$$

$$\hat{p}_1 = \frac{Y_{1,1} + Y_{2,1}}{n_1 + n_2}, \hat{p}_2 = \frac{Y_{1,2} + Y_{2,2}}{n_1 + n_2}, \dots, \hat{p}_k = \frac{Y_{1,k} + Y_{2,k}}{n_1 + n_2}$$

k-1: only k-1 of the proportions need to be estimated.

$$\hat{p}_k = 1 - \hat{p}_1 - \hat{p}_2 - \cdots - \hat{p}_{k-1}$$

$$\sum_{i=1}^{2} \sum_{j=1}^{k} \frac{\left(y_{i,j} - n_i \hat{p}_j\right)^2}{n_i \hat{p}_j} \sim \chi^2\left(\boldsymbol{2(k-1) - (k-1)}\right) = \boldsymbol{\chi^2(k-1)}$$

The Contingency Table (summarize)

	A_1	A_2	A_3	$\cdots$	A_k	Total
First experiment	$Y_{1,1}$ (Observed) $n_1\hat{p}_1$ (expected)	$Y_{1,2}$ $n_1\hat{p}_2$	$Y_{1,3}$ $n_1\hat{p}_3$	$\cdots$	$Y_{1,k}$ $n_1\hat{p}_k$	n_1
Second experiment	$Y_{2,1}$ $n_2\hat{p}_1$	$Y_{2,2}$ $n_2\hat{p}_2$	$Y_{2,3}$ $n_3\hat{p}_3$	$\cdots$	$Y_{2,k}$ $n_2\hat{p}_k$	n_2
Total	$Y_{1,1}+Y_{2,1}$ $\left(\hat{p}_1=\frac{Y_{1,1}+Y_{2,1}}{n_1+n_2}\right)$	$Y_{1,2}+Y_{2,2}$ $\left(\hat{p}_2=\frac{Y_{1,2}+Y_{2,2}}{n_1+n_2}\right)$	$Y_{1,3}+Y_{2,3}$ $\left(\hat{p}_3=\frac{Y_{1,3}+Y_{2,3}}{n_1+n_2}\right)$		$Y_{1,k}+Y_{2,k}$ $\left(\hat{p}_k=\frac{Y_{1,k}+Y_{2,k}}{n_1+n_2}\right)$	n_1+n_2

$$\chi^2=\sum_{i=1}^{2}\sum_{j=1}^{k}\frac{\left(y_{i,j}-n_i\hat{p}_j\right)^2}{n_i\hat{p}_j}\sim \boldsymbol{\chi^2(k-1)}$$

Critical (Rejection) region

$$\chi^2>\chi^2_\alpha(k-1)$$

at significance level α

p-value

$$P(\chi^2(k-1)>\chi^2)$$

Example

Example 9.2-1 To test two methods of instruction, 50 students are selected at random from each of two groups. At the end of the instruction period, each student is assigned a grade (A, B, C, D, or F) by an evaluating team. The data are recorded as follows:

	Grade					
	A	B	C	D	F	Totals
Group I	8	13	16	10	3	50
Group II	4	9	14	16	7	50

Example:

$\hat{p}_A = \frac{8+4}{100} = 0.12$ $\hat{p}_B = \frac{13+9}{100} = 0.22$

$\hat{p}_C = \frac{16+14}{100} = 0.30$ $\hat{p}_D = \frac{10+16}{100} = 0.26$

$\hat{p}_F = \frac{3+7}{100} = 0.1$

Example

	Grade					Total
	A	B	C	D	F	
Group I	8 (6)	13 (11)	16 (15)	10 (13)	3 (5)	50
Group II	4 (6)	9 (11)	14 (15)	16 (13)	7 (5)	50
Total	12	22	30	26	10	100

expected: $50 \times \hat{p}_A = 50 \times 0.12 = 6$

$50 \times \hat{p}_B = 11$

$\vdots$

$50 \times \hat{p}_F = 5$

$$\chi^2 = \sum_{i=1}^{2}\sum_{j=1}^{5} \frac{\left(y_{i,j} - n_i\hat{p}_j\right)^2}{n_i\hat{p}_j} = \frac{(8-6)^2}{6} + \frac{(13-11)^2}{11} + \frac{(16-15)^2}{15} + \frac{(10-13)^2}{13} + \frac{(3-5)^2}{5}$$

$$+ \frac{(4-6)^2}{6} + \frac{(9-11)^2}{11} + \frac{(14-15)^2}{15} + \frac{(16-13)^2}{13} + \frac{(7-5)^2}{5}$$

$$= \frac{4}{6} + \frac{4}{11} + \frac{1}{15} + \frac{9}{13} + \frac{4}{5} + \frac{4}{6} + \frac{4}{11} + \frac{1}{15} + \frac{9}{13} + \frac{4}{5} = 5.18. < 9.488 = \chi^2_{0.05}(4)$$

$\sim \chi^2(k-1) = \chi^2(5-1)$

Under H0

Fail to reject the null hypothesis

Example

- We can also test whether two continuous distributions are equal

$$H_0\text{: } F(x) = G(x) \text{ for all } x$$

- Partition the real line into k disjoint sets $A_1, A_2, \dots, A_k$

$$p_{1,i} = P(U \in A_i), U \sim F; \quad p_{2,i} = P(V \in A_i), V \sim G, \qquad i = 1, \dots, k$$

- If $F(x) = G(x), \forall x$, then $p_{1,i} = p_{2,i}, i = 1, \dots, k$

$$H_0'\text{: } p_{1,i} = p_{2,i}, \qquad i = 1,2 \dots, k$$

Example

Example 9.2-2

Select, at random, 20 cars of each of two comparable major-brand models. All 40 cars are submitted to accelerated life testing; that is, they are driven many miles over very poor roads in a short time, and their failure times (in weeks) are recorded as follows:

Brand U:	25	31	20	42	39	19	35	36	44	26
	38	31	29	41	43	36	28	31	25	38
Brand V:	28	17	33	25	31	21	16	19	31	27
	23	19	25	22	29	32	24	20	34	26

If we use 23.5, 28.5, and 34.5 as dividing marks, we note that exactly one fourth of the 40 cars fall into each of the resulting four sets. Thus, the data can be summarized as follows:

	A_1	A_2	A_3	A_4	Totals
Brand U	2	4	4	10	20
Brand V	8	6	6	0	20

Example:

	Categories				
	A_1	A_2	A_3	A_4	Total
U	2 (5)	4 (5)	4 (5)	10 (5)	20
V	8 (5)	6 (5)	6 (5)	0 (5)	20
Total	10	10	10	10	
	$\hat{P}_{A_1} = \frac{1}{4}$	$\hat{P}_{A_2} = \frac{1}{4}$	$\hat{P}_{A_3} = \frac{1}{4}$	$\hat{P}_{A_4} = \frac{1}{4}$	

$$\chi^2 = \frac{(2-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(10-5)^2}{5}$$

$$+ \frac{(8-5)^2}{5} + \frac{(6-5)^2}{5} + \frac{(6-5)^2}{5} + \frac{(0-5)^2}{5}$$

$$= 14.4 > \chi^2_{0.05}(3) = 7.815$$

Here degree of freedom = $k-1 = 4-1 = 3$

$\Rightarrow$ Reject H_0

Also, p-value = $P(\chi^2(3) > 14.4) \approx 0.0024 < 0.05$

h groups and k categories

- h (e.g., $h = 2$ in previous examples) independent experiments, each experiment has *k* mutually exclusive and exhaustive outcomes, $A_1, A_2, \dots, A_k$

	A_1	A_2	A_3	$\cdots$	A_k
First experiment	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$\cdots$	$p_{1,k}$
Second experiment	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$\cdots$	$p_{2,k}$
……	…	…	…	…	…
h-th experiment	$p_{h,1}$	$p_{h,2}$	$p_{h,3}$	$\cdots$	$p_{h,k}$

multinomial distributions

$$p_{1,1} + p_{1,2} + \cdots p_{1,k} = 1$$

$$\vdots$$

$$p_{h,1} + p_{h,2} + \cdots p_{h,k} = 1$$

- Test homogeneity (the corresponding probabilities associated with the *h* independent experiments are equal)

$$H_0: \ p_{i,j} = p_j, \forall i = 1,2, \dots, h, \qquad j = 1,2, \dots, k$$

$h \times k$ Contingency Table

	A_1	A_2	A_3	$\cdots$	A_k	Total
First	$Y_{1,1}$ (Observed) $n_1\hat{p}_1$ (expected)	$Y_{1,2}$ $n_1\hat{p}_2$	$Y_{1,3}$ $n_1\hat{p}_3$	$\cdots$	$Y_{1,k}$ $n_1\hat{p}_k$	n_1
Second	$Y_{2,1}$ $n_2\hat{p}_1$	$Y_{2,2}$ $n_2\hat{p}_2$	$Y_{2,3}$ $n_3\hat{p}_3$	$\cdots$	$Y_{2,k}$ $n_2\hat{p}_k$	n_2
…	…	…	…		…	…
h-th	$Y_{h,1}$ $n_h\hat{p}_1$	$Y_{h,2}$ $n_h\hat{p}_2$	$Y_{h,3}$ $n_h\hat{p}_3$		$Y_{h,k}$ $n_h\hat{p}_k$	n_k
Total	$Y_{1,1} + \cdots + Y_{h,1}$ $\hat{p}_1 = \frac{Y_{1,1} + \cdots + Y_{h,1}}{n_1 + \cdots + n_h}$	$Y_{1,2} + \cdots + Y_{h,2}$ $\hat{p}_2 = \frac{Y_{1,2} + \cdots + Y_{h,2}}{n_1 + \cdots + n_h}$	$Y_{1,3} + \cdots + Y_{h,3}$ $\hat{p}_3 = \frac{Y_{1,3} + \cdots + Y_{h,3}}{n_1 + \cdots + n_h}$		$Y_{1,k} + \cdots + Y_{h,k}$ $\hat{p}_k = \frac{Y_{1,k} + \cdots + Y_{h,k}}{n_1 + \cdots + n_h}$	$n_1 + \cdots + n_h$

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k} \frac{\left(y_{i,j} - n_i\hat{p}_j\right)^2}{n_i\hat{p}_j} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha((h-1)(k-1))$$

p-value

$$P(\chi^2_\alpha((h-1)(k-1)) > \chi^2)$$

at significance level α

Example

- Suppose we have the following 3×2 Contingency Table, test the homogeneity

Group	L	H	Totals
1	9	1	10
2	5	5	10
3	1	9	10
Totals	15	15	30

Example

Example.

Group	Category L	Category H	Total
1	9 (5)	1 (5)	10
2	5 (5)	5 (5)	10
3	1 (5)	9 (5)	10
Total	15	15	30

$\hat{P}_L = \frac{1}{2}$ $\hat{P}_H = \frac{1}{2}$

expected. $10 \times \hat{P}_L = 10 \times \hat{P}_H = 5$

Example

chi-square statistics

$$\chi^2 = \frac{(9-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(9-5)^2}{5} = 12.5$$

degree of freedom: $(h-1)(k-1) = (3-1)\times(2-1) = 2$

critical region: $\{\chi^2 > \chi^2_{0.05}(2) = 5.99\}$.

Reject H_0.

p-value $= P(\chi^2(2) > 12.5) = 0.002 < 0.05$

Testing Independence

Experiment with two attributes

- Suppose a random experiment results in an outcome that can be classified by *two different attributes*
- The first attribute is assigned to one and only one of *h* mutually exclusive and exhaustive event—say $A_1, A_2, \dots, A_h$ —and the second attribute falls into one and only one of *k* mutually exclusive and exhaustive events—say $B_1, B_2, \dots, B_k$

$$p_{i,j} = P(A_i \cap B_j), \qquad i = 1,2,\dots,h, j = 1,2,\dots,k$$

- Repeat the experiment *n* times, and let $Y_{i,j}$ denotes the number of events $A_i \cap B_j$, then the chi-square test says

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k} \frac{(y_{i,j} - np_{i,j})^2}{np_{i,j}} \sim \boldsymbol{\chi^2(hk-1)}$$

Testing Independence

- Suppose we wish to test the hypothesis that the attributes *A* and *B* are independent, i.e.,

$$p_{i,j} = P(A_i \cap B_j) = P(A_i)P(B_j), \qquad i = 1,2,\dots,h, j = 1,2,\dots,k$$

- Denote $p_{i,\cdot} = \sum_{j=1}^{k} p_{i,j}$ as the marginal distribution for *A*
- Denote $p_{\cdot,j} = \sum_{i=1}^{h} p_{i,j}$ as the marginal distribution for *B*
- If A and B are independent, then it means

$$p_{i,j} = p_{i,\cdot} \times p_{\cdot,j}, \qquad i = 1,2,\dots,h, j = 1,2,\dots,k$$

$$H_0: \; p_{i,j} = p_{i,\cdot} \times p_{\cdot,j}, \qquad \forall\, i = 1,2,\dots,h, \forall\, j = 1,2,\dots,k$$

Testing Independence

- However, if $p_{i,\cdot}, p_{\cdot,j}$ are **unknown**, under H_0, we estimate $p_{i,\cdot}, p_{\cdot,j}$

$$\hat{p}_{i,\cdot} = \frac{\sum_{j=1}^{k} Y_{i,j}}{n}, \qquad \hat{p}_{\cdot,j} = \frac{\sum_{i=1}^{h} Y_{i,j}}{n}$$

h-1: only h-1 of the proportions need to be estimated.

$$\hat{p}_{h,\cdot} = 1 - \hat{p}_{1,\cdot} - \hat{p}_{2,\cdot} - \cdots - \hat{p}_{k-1,\cdot}$$

k-1: only k-1 of the proportions need to be estimated.

$$\hat{p}_{\cdot,k} = 1 - \hat{p}_{\cdot,1} - \hat{p}_{\cdot,2} - \cdots - \hat{p}_{\cdot,k-1}$$

$$H_0: \; p_{i,j} = \hat{p}_{i,\cdot}\hat{p}_{\cdot,j}, \qquad i = 1, \ldots, h, j = 1, \ldots, k$$

$$\sum_{i=1}^{h}\sum_{j=1}^{k} \frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} \sim \chi^2\left(\boldsymbol{kh - 1 - (h-1) - (k-1)}\right) = \boldsymbol{\chi^2\left((k-1)(h-1)\right)}$$

Number of parameters that have been estimated

$h \times k$ Contingency Table

	B_1	B_2	$\cdots$	B_k	Total
A_1	$Y_{1,1}$ (Observed) $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,1}$ (expected)	$Y_{1,2}$ $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{1,k}$ $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,k}$	$Y_{1,\cdot}$ $\hat{p}_{1,\cdot} = Y_{1,\cdot}/n$
A_2	$Y_{2,1}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,1}$	$Y_{2,2}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{2,k}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,k}$	$Y_{2,\cdot}$ $\hat{p}_{1,\cdot} = Y_{2,\cdot}/n$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
A_h	$Y_{h,1}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,1}$	$Y_{h,2}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{h,k}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,k}$	$Y_{h,\cdot}$ $\hat{p}_{h,\cdot} = Y_{h,\cdot}/n$
Total	$Y_{\cdot,1}$ $\hat{p}_{\cdot,1} = Y_{\cdot,1}/n$	$Y_{\cdot,2}$ $\hat{p}_{\cdot,2} = Y_{\cdot,2}/n$		$Y_{\cdot,k}$ $\hat{p}_{\cdot,k} = Y_{\cdot,k}/n$	n

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k} \frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha((h-1)(k-1))$$

p-value

$$P(\chi^2_\alpha((h-1)(k-1)) > \chi^2)$$

Example

Example 9.2-3

The 400 undergraduate students in a random sample at the University of Iowa were classified according to the college in which the students were enrolled and according to their gender. The results are recorded in Table 9.2-1, called a $k \times h$ **contingency table**, where, in this case, $k = 2$ and $h = 5$. (Do not be concerned about the numbers in parentheses at this point.) Incidentally, these data do actually reflect the composition of the undergraduate colleges at Iowa, but they were modified a little to make the computations easier in this example.

Table 9.2-1 Undergraduates at the University of Iowa

	College					
Gender	Business	Engineering	Liberal Arts	Nursing	Pharmacy	Totals
Male	21	16	145	2	6	190
	(16.625)	(9.5)	(152)	(7.125)	(4.75)	
Female	14	4	175	13	4	210
	(18.375)	(10.5)	(168)	(7.875)	(5.25)	
Totals	35	20	320	15	10	400

Example

We desire to test the null hypothesis H_0: $p_{ij} = p_{i\cdot}p_{\cdot j}$, $i = 1, 2$ and $j = 1, 2, 3, 4, 5$, that the college in which a student enrolls is independent of the gender of that student. Under H_0, estimates of the probabilities are

Under significance level 0.01

$$\widehat{p}_{1\cdot} = \frac{190}{400} = 0.475 \qquad \text{and} \qquad \widehat{p}_{2\cdot} = \frac{210}{400} = 0.525$$

and

$$\widehat{p}_{\cdot 1} = \frac{35}{400} = 0.0875,\ \widehat{p}_{\cdot 2} = 0.05,\ \widehat{p}_{\cdot 3} = 0.8,\ \widehat{p}_{\cdot 4} = 0.0375,\ \widehat{p}_{\cdot 5} = 0.025.$$

$$\sum_{i=1}^{2}\sum_{j=1}^{5}\frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} = \frac{(21-16.625)^2}{16.625} + \frac{(14-18.375)^2}{18.375} + \cdots + \frac{(4-5.25)^2}{5.25}$$

$$= 1.15 + 1.04 + 4.45 + 4.02 + 0.32 + 0.29 + 3.69$$

$$+ 3.34 + 0.33 + 0.30 = 18.93. > 13.28 = \chi^2_{0.01}(4)$$

Reject null hypothesis

Example

Example 9.2-5

Two groups of workers have the same qualifications for a particular type of work. Their experience in salaries is summarized by the following 2×5 contingency table, in which the upper bound of each salary range is not included in that listing:

	Salary (Thousands of Dollars)					
Group	27–29	29–31	31–33	33–35	35 and over	Totals
1	6	11	16	14	13	60
2	5	9	8	6	2	30
Totals	11	20	24	20	15	90

Example

To test whether the group assignment and the salaries seem to be independent with these data at the $\alpha = 0.05$ significance level, we compute

$$q = \frac{[6 - 90(60/90)(11/90)]^2}{90(60/90)(11/90)} + \cdots + \frac{[2 - 90(30/90)(15/90)]^2}{90(30/90)(15/90)}$$

$$= 4.752 \ < \ 9.488 \ = \ \chi^2_{0.05}(4).$$

Also, the p-value is 0.314. Hence, with these limited data, group assignment and salaries seem to be independent. ■

Summary

- **Homogeneity Tests**
  - h groups of experiments, and k categories
  - Test whether the multinomial distribution for each group is the same

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k}\frac{\left(y_{i,j} - n_i\hat{p}_j\right)^2}{n_i\hat{p}_j} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

- **Independence Tests**
  - Experiments has two attributes each with h and k possible outcomes, test whether they are independent

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k}\frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$